SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 10%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

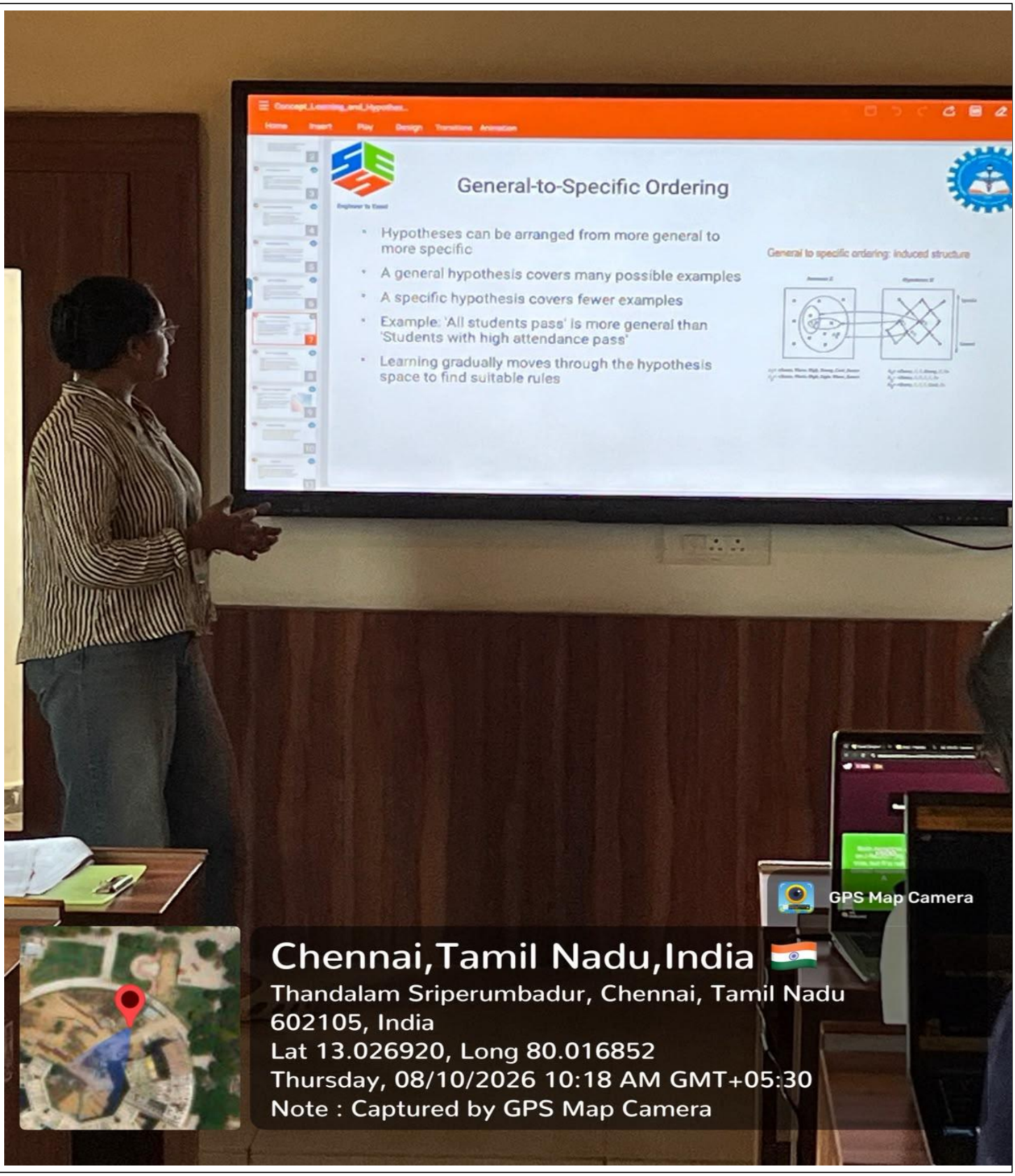
TECHNICAL PRESENTATION

Code of Conduct:

I K.KAVIYA 192521262 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

KAVIYA K


General-to-Specific Ordering
Hypotheses can be arranged from more general to more specific
A general hypothesis covers many possible examples
A specific hypothesis covers fewer examples
Example: 'All students pass' is more general than 'Students with high attendance pass'
Learning gradually moves through the hypothesis space to find suitable rules
General to specific ordering: induced structure
GPS Map Camera
Chennai,Tamil Nadu,India
Thandalam Sriperumbadur, Chennai, Tamil Nadu 602105, India
Lat 13.026920, Long 80.016852
Thursday, 08/10/2026 10:18 AM GMT+05:30
Note : Captured by GPS Map Camera

Why is Inductive Bias Important?

Inductive bias helps the algorithm to:

- Learn from **limited data**
- Reduce the number of possible hypotheses
- Select a suitable solution
- Make predictions on **unseen examples**
- Improve generalization

8. Inductive Bias in Learning Algorithms: Define inductive bias and explain its significance in machine learning. Discuss different types of bias such as restriction bias and preference bias. Explain how inductive bias helps a learning algorithm generalize from limited data and influences hypothesis selection in concept learning.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**35**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**25**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**10**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand